

## **Project Title**

# **E-Commerce Sales Analysis and Performance Dashboard using Power BI**

### **Introduction**

The objective of this project is to analyse e-commerce sales data using Power BI and generate meaningful business insights through DAX calculations and interactive visualizations. The project focuses on evaluating sales performance, profitability, sales targets, and regional trends to support data-driven decision-making.

### **Dataset Description**

The analysis was performed using three datasets:

| <b>DATASET</b> | <b>DESCRIPTION</b>                                                                  |
|----------------|-------------------------------------------------------------------------------------|
| List of Orders | Contains Order details, Customer Information, City, State, Sales amount and Profit. |
| Order Details  | Contains category, sub-category, quantity, amount, and profit information.          |
| Sales Target   | Contains category-wise monthly sales targets.                                       |

### **DAX Calculations**

The following calculated columns and measures were created: **Calculated**

#### **Columns**

Category Type

Revenue per Order

Sales Category

#### **Measures**

Order Count

Average Profit in Delhi

Year-to-Date (YTD) Sales

These calculations helped derive important business metrics for analysis and reporting.

## Data Visualization and Analysis

The dashboard includes various visualizations to analyse sales performance and business trends effectively.

\*Sales Target Achievement by Category: Used to compare actual sales against predefined sales targets and identify categories that met or missed expectations.

\* Profit Margin by Sub-Category: Helps determine which sub-categories generate the highest profit margins and contribute most to business profitability.

\* Monthly Sales Trend: Displays sales performance over different months, making it easier to identify growth patterns and seasonal variations.

\* Profit vs Quantity Analysis: Examines the relationship between the quantity of products sold and the profit earned from those sales.

\* Sales and Target KPI Cards: Provides a quick overview of key business metrics such as total sales and sales targets.

\* Sales Performance Matrix: Enables comparison of sales performance across categories and different time periods.

\* Geographical Sales Analysis: Visualizes sales distribution across cities and states to identify high-performing regions.

\* Sales Distribution Tree map: Shows the contribution of each sub-category to overall sales, helping identify top-performing products.

\* Order Count Analysis: Analysis the number of orders across different states to understand customer distribution and market reach.

## Dashboard Features

Interactive and user-friendly dashboard

Clear visualization of sales performance

Business KPI monitoring through cards and measures

Regional sales analysis using maps

Trend analysis using line charts

Comparative analysis using matrix and scatter charts

## **Key Insights**

Electronics category generated the highest revenue.

Certain sub-categories contributed significantly to overall profit.

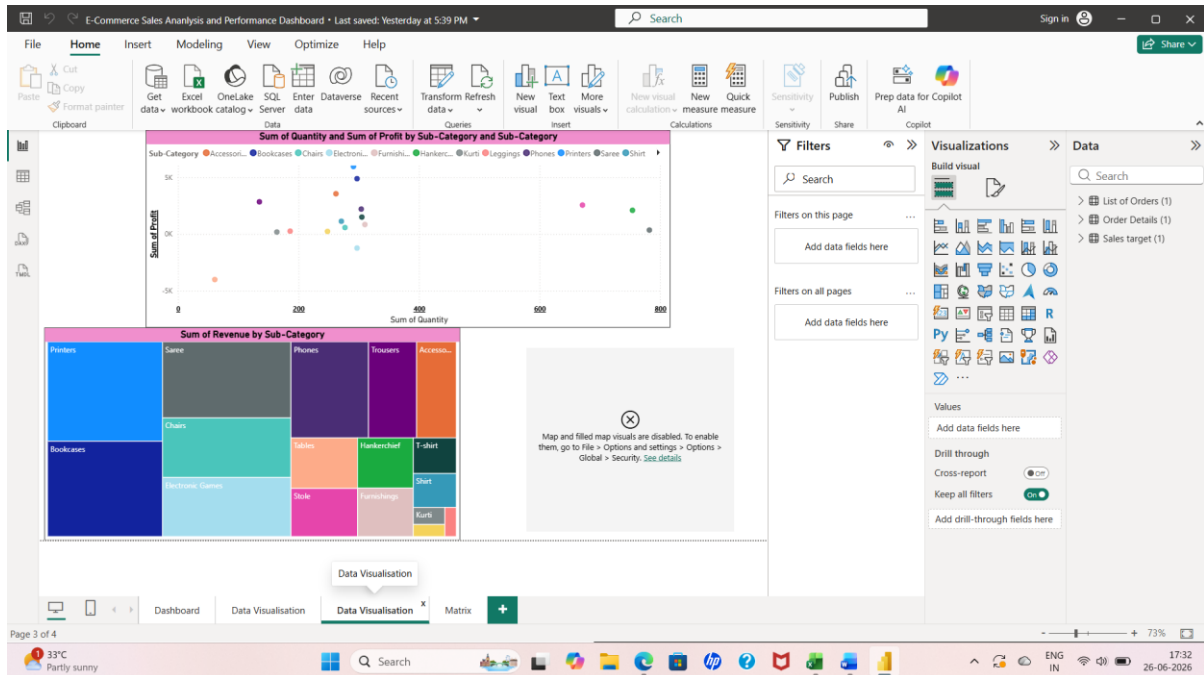
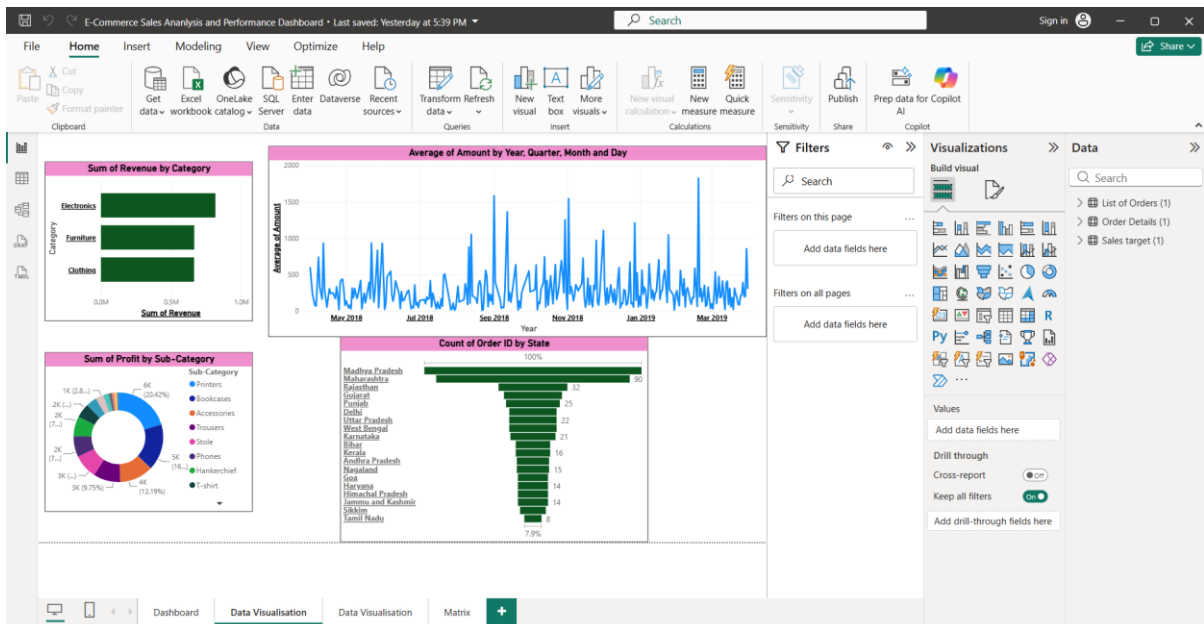
Sales performance varied across different months and regions.

Comparison with sales targets helped identify performance gaps.

Geographic analysis revealed major sales-contributing cities and states.

## **Conclusion**

This project successfully utilized Power BI, DAX calculations, and data visualization techniques to transform raw e-commerce data into actionable business insights. The dashboard enables effective monitoring of sales performance, profitability, and target achievement, helping organizations make informed business decisions and improve overall performance.



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| Year        | 2018           |               | 2019           |               | Total          |               |
|-------------|----------------|---------------|----------------|---------------|----------------|---------------|
|             | Sum of Revenue | Sum of Target | Sum of Revenue | Sum of Target | Sum of Revenue | Sum of Target |
| Clothing    | 455537         | 435900        | 208985         | 435900        | 664522         | 435900        |
| Electronics | 507122         | 435900        | 309461         | 435900        | 816583         | 435900        |
| Furniture   | 347612         | 435900        | 318153         | 435900        | 665765         | 435900        |
| Total       | 1310271        | 435900        | 836999         | 435900        | 2146870        | 435900        |

Matrix

Filters

Visualizations

Data

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Total order count  
500

Average profit Delhi  
43.31

YTD Sales  
158800

Dashboard

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